

Introduction:

Task 4 involved the final stage of the project: transforming the Task 3 Smart Automation System into a fully functional, cloud-connected IoT device. The primary objective was to establish reliable, two-way communication with a free cloud IoT platform (Blynk) to enable **real-time monitoring of sensor data** and **remote notification** of intrusion events from any geographical location.

This task validated the ability to bridge local embedded systems (ESP32) with global network services, achieving the core goal of an Internet of Things application.

IoT System Architecture and Data Flow

The final system operates on a three-tier architecture: the Device Layer, the Cloud/Platform Layer, and the Application Layer.

Three-Tier Architecture

1. **Device Layer (ESP32):** Reads the PIR sensor state, executes the alarm logic, and packages the data.
2. **Cloud Layer (Blynk Server):** Authenticates the device, receives data streams, stores history, and routes notifications.
3. **Application Layer (Blynk Mobile/Web App):** Provides the customizable dashboard for visualization and remote interaction.

The system uses **Virtual Pins** as abstract communication channels between the device and the cloud, simplifying data transfer regardless of physical wiring.

Connection Management:

The following code elements ensure robust and authenticated connectivity:

- **Authentication:** The template ID, template name, and unique authentication token (BLYNK_AUTH_TOKEN) are defined at the top of the sketch, allowing the Blynk.begin() function to establish a secure, specific connection to the designated device instance on the Blynk server.
- **Persistent Link:** The non-blocking Blynk.run() function is placed inside the loop(). This command continuously checks for incoming data from the cloud (commands, settings) and processes any outgoing data streams, maintaining the open connection with minimal overhead.

B. Sensor Data Handling

The state of the alarm flag (isAlarmActive) from the Task 3 logic is used directly to trigger cloud updates:

- **Trigger Event:** Data transmission only occurs when the alarm state **changes** (e.g., transitions from OFF to ON, or from ON back to OFF after the 2.5-second cooldown).
- **Immediate Update:** When isAlarmActive becomes true, the code instantly calls Blynk.virtualWrite(V0, 1) and sends a descriptive alert message to V1, ensuring zero latency in reporting an intrusion.

Dashboard:

